

Due: Tuesday, April 1st, before class.

Name:

Collaborators:

Document your work and note any collaborators. In your write-up, please neatly document any code you have written, alongside its output, and including any necessary commentary to answer the question. Please use an `.ipynb` notebook for any code, include commentary in markdown code blocks, and print the `.ipynb` notebook to PDF to attach to your submission. The L^AT_EXsource for this file may be helpful: <https://www.overleaf.com/read/wvsrgcbwmcwc#9a682a>.

Please make sure to label the parts of your assignment! Use markdown blocks in your Julia notebook and make headings using #, ##, and ###.

There are many important *activator-inhibitor* phenomena, in which two components of a system essentially compete to pull it in different directions, leading to oscillatory behavior. A classic example is the *FitzHugh-Nagumo* model of a neuron:

$$\begin{aligned}\frac{dv}{dt} &= v(a - v)(v - 1) - w + \beta, \\ \frac{dw}{dt} &= bv - cw,\end{aligned}\tag{1}$$

where $0 < a < 1$, and b , c , and β are positive constants. In physically realistic models, v stands for an activation potential, and the peaks in v correspond to “firing” a signal. The parameter β is analogous to an external current enhancing the growth of v , and it affects the period of the oscillations.

A steady state of the ODE occurs when $v'(t) = w'(t) = 0$. These conditions imply a nonlinear system of two equations for the two variables v and w .

Goal 1: Let $a = 1$ and $c = 4$. Starting from the values $v = 1$, $w = 0$, calculate by hand the symbolic result of taking one Newton step to solve the nonlinear system. Your answer will be in terms of b and β . (*Note:* It would be trivial to solve for $w = bv/c$, eliminate w , and obtain a single equation in only v . But for the purposes of this problem, leave it as two equations in two variables.)

Procedure: Perform the following steps.

- (a) Define (write down) the function $\mathbf{f}(\mathbf{x})$ with $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $\mathbf{x} = \begin{pmatrix} v \\ w \end{pmatrix}$ whose root, $\mathbf{r}$ with $\mathbf{f}(\mathbf{r}) = \mathbf{0}$, provides a steady state of the ODE.

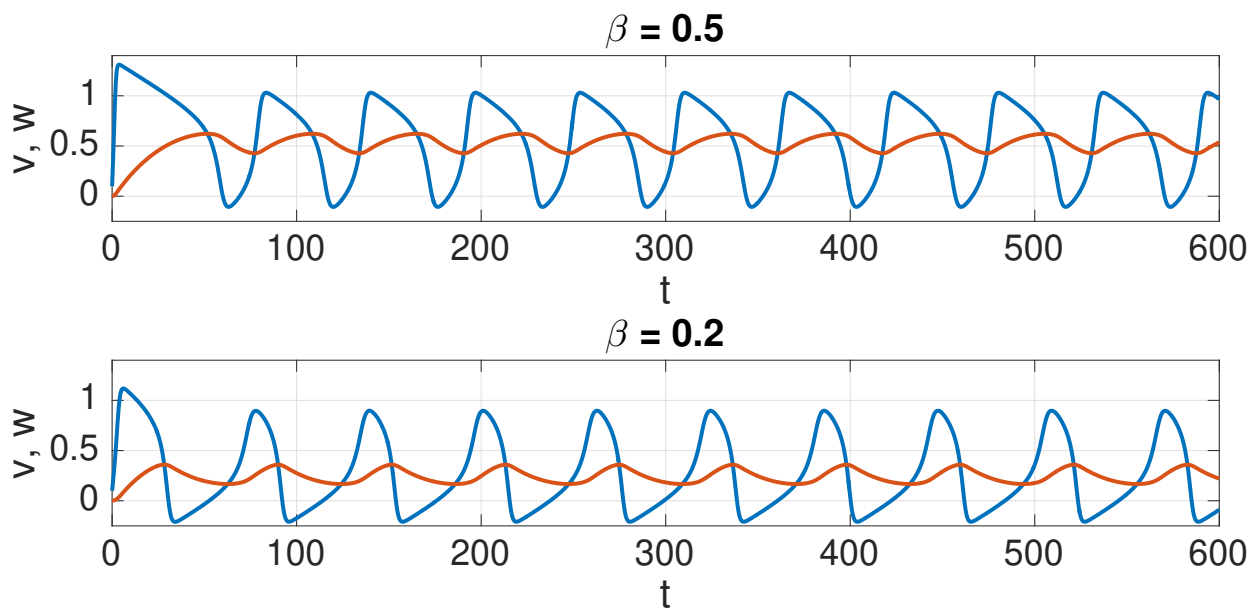


Figure 1: Solutions of the system (1) using the values in (2) and two different values of β .

- (b) Calculate the Jacobian matrix, $\mathbf{J}(\mathbf{x})$.
- (c) Evaluate the function $\mathbf{f}(\mathbf{x})$ and the Jacobian $\mathbf{J}(\mathbf{x})$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- (d) Compute the output of one step of Newton's method; that is, the values of v and w after the Newton step.

You can see solutions of the model as an initial-value problem for two different values of β in Figure 1, for which

$$a = 0.25, \quad b = c = 0.02, \quad v(0) = 0.1, \quad w(0) = 0. \quad (2)$$

Goal 2: Recreate Figure 1 using the single-neuron model (1) and the parameters in (2). (You do not need to worry about details such as line width, font size, axes size, etc.) Then, on one other graph, plot the two solutions in the (v, w) phase plane.

Procedure: Perform the following steps.

- (a) Implement a function `fhn` for the model (1) as we did in class.
- (b) Recreate the first plot in Figure 1, plotting the solution of the FHN model using $\beta = 0.5$ and the other parameters given in (2). Calculate the solution using the `DifferentialEquations` package as in class.

- (c) Plot this solution in the (v, w) phase plane.
- (d) Recreate the second plot in Figure 1, plotting the solution of the FHN model using $\beta = 0.2$ and the other parameters given in (2).
- (e) Plot this solution in the (v, w) phase plane.

From Figure 1 (top half) we can estimate the period of oscillation when $\beta = 0.5$. Imagine drawing a horizontal line in Figure 1 with $v = 1/2$. For each oscillation, this line makes two intersections with the graph of $v(t)$ in blue: one when v is increasing, and another when v is decreasing. Hence we can define one period of the oscillation as the time between two similar kinds of intersections (i.e., both increasing or both decreasing).

We can extend the idea to get much more precision than a simple graphical estimate allows. Note that whichever kind of intersection we pick (increasing or decreasing), there is a particular value of w associated with it.

We can define the period T via the conditions $v(0) = v(T) = 1/2$, $w(0) = w(T)$. If we let the initial conditions be $v(0) = 1/2$, $w(0) = \gamma$, then we can solve the model over $[0, T]$ and check the equations $v(T) = 1/2$, $w(T) = \gamma$, which will be valid only when γ and T are chosen correctly.

In other words, the required equations for $v(T)$ and $w(t)$ define a system of two equations $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ for the unknowns $x_1 = \gamma$ and $x_2 = T$.

Goal 3: Find the oscillation period with $a = 0.25$, $b = c = 0.02$, and $\beta = 0.5$ by using `levenberg` from our class notes (make sure to cite that you are using the implementation from class) on the system of equations described above.

As always, you must provide `levenberg` with the function $\mathbf{f}(\mathbf{x})$. You can find a good starting value for `levenberg` using the graph you made for Objective 2.

Procedure: Perform the following steps.

- (a) Define the appropriate function $\mathbf{f}$. Note that your function will include redefining in the initial values and solution interval for the ODE problem in terms of γ and T , and will solve the ODE for $v(t)$ and $w(t)$, which then will be evaluated in the output of $\mathbf{f}$.
- (b) Identify good initial guesses for γ and T using the plot for $\beta = 0.5$ in Figure 1 above.
- (c) Solve for γ and T using `levenberg` (make sure to set the max number of iterations in the code to at least 400) initialized at your guesses from part (b).
- (d) Take γ and T from the last output from `levenberg`, and solve the ODE problem with these values of γ and T . Create a plot of v, w as in Figure 1. Explain where you see γ and T in this plot.

Some oscillators exhibit an interesting phenomenon of synchronization. If the oscillators are coupled together, even weakly, they can adjust their individual natural frequencies to oscillate together. This can occur with pendulums, fireflies¹, and neurons. To extend our equations to two neurons, we use

$$\begin{aligned}
 \frac{dv_1}{dt} &= v_1(a - v_1)(v_1 - 1) - w_1 + \beta_1 + \alpha v_2, \\
 \frac{dv_2}{dt} &= v_2(a - v_2)(v_2 - 1) - w_2 + \beta_2 + \alpha v_1, \\
 \frac{dw_1}{dt} &= bv_1 - cw_1, \\
 \frac{dw_2}{dt} &= bv_2 - cw_2,
 \end{aligned} \tag{3}$$

Here, the excitation of v_i occurs in response to the firing of the other neuron, with a strength expressed as $\alpha > 0$. Larger values of α mean stronger coupling.

Goal 4: Implement the model (3), using $\beta_1 = 0.5$, $\beta_2 = 0.2$, $v_1(0) = v_2(0) = 0.25$, $w_1(0) = w_2(0) = 0$, and a , b , and c as given in (2). Make a 2×1 subplot grid. In the top axes plot $v_1(t)$ and $v_2(t)$ over $0 \leq t \leq 1000$ for $\alpha = 0.002$, and in the bottom axes make the same plot for the solution with $\alpha = 0.05$. It should be clear that the top pair never synchronize while the bottom pair quickly reach the same frequency.

Procedure: Perform the following steps.

- (a) Implement a function `fhn2` for the model (3) as we did in class.
- (b) Calculate the solution to the system (3) for $\alpha = 0.002$, $\beta_1 = 0.5$, $\beta_2 = 0.2$, $v_1(0) = v_2(0) = 0.25$, $w_1(0) = w_2(0) = 0$, and a , b , and c as given in (2). Plot $v_1(t)$ and $v_2(t)$ over $0 \leq t \leq 1000$.
- (c) Repeat the step above with $\alpha = 0.05$.

¹See https://youtu.be/t-_VPRCtiUg?t=485, for example.